Fund Management Capstone

Final Report

Aggressive-Growth Retirement Fund Redesign

BFIN-465 Financial Analytics with Python
Spring 2026

Manager role:	Newly Hired Equity Fund Manager
Decision date:	January 1, 2020
Evaluation window:	2020-01-01 through 2025-12-31
Benchmark:	S&P 500 Index (SPY)

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Contents

1 Executive Summary **1**

2 Project Background and Investment Mandate **1**

3 Data and Methodology **2**

4 Inherited Fund Diagnosis **2**

5 Candidate Search and Final Revised Portfolio **3**

6 Backtest Results **5**

7 Active Rule Selection **6**

8 Factor Analysis: CAPM and Fama–French 3-Factor Model **7**

8.1 Full-sample factor results . . . 7

8.2 Rolling 3-year factor exposures . . . 7

8.3 Stock-level FF3 contribution . . . 8

9 Scenario and Stress Testing **9**

9.1 Historical realized stress tests . . . 10

9.2 Deterministic portfolio shocks . . . 10

9.3 Factor stress scenarios . . . 10

9.4 Bootstrap distribution of terminal returns . . . 11

10 Limitations **11**

11 Final Recommendation **12**

11.1 Theme of the revised fund . . . 12

11.2 Keep, drop, and add decisions . . . 13

11.3 How the revised fund is implemented . . . 13

11.4 Why this recommendation is preferred . . . 13

1 Executive Summary

Recommendation. Replace the inherited 10-stock fund with an aggressive-growth revised portfolio and implement it with **monthly equal-weight rebalancing**. The revised active implementation captures substantial upside while controlling the winner-driven concentration that appears in both the inherited fund and the revised static portfolio.

The inherited fund delivered strong historical performance, but by the January 2020 takeover date it had drifted away from its original equal-weight design. The fund became highly dependent on a small set of winners, especially large growth-oriented technology and platform companies. The redesign problem is therefore not how to rescue a failed portfolio; it is how to preserve the strongest inherited compounders while making the portfolio more intentional for a long-horizon investor.

The revised mandate is to build a growth-oriented retirement equity fund for an investor in their 30s. This investor can tolerate meaningful equity risk, but the portfolio should take that risk deliberately rather than through unmanaged concentration. The revised fund keeps five inherited holdings—AAPL, AMZN, BRK-B, GOOG, and MSFT—and adds five candidates: NVDA, AVGO, ADBE, COST, and UNH. It drops CMCSA, KO, ORCL, WFC, and XOM.

Out of sample from 2020 through 2025, the revised portfolios outperform both the Legacy Fund and the S&P 500 Index. The Revised Static Fund produces the highest terminal wealth, but it does so by allowing large winner-driven concentration to form. The Revised Active Fund gives up some of that upside but better controls portfolio drift. For the final recommendation, the active implementation is preferred because it provides a more disciplined balance between long-run growth and portfolio risk control.

2 Project Background and Investment Mandate

The case begins with an inherited equity-only fund launched on **January 1, 2010** with $1,000,000 invested equally across 10 stocks: AAPL, AMZN, BRK-B, CMCSA, GOOG, KO, MSFT, ORCL, WFC, and XOM. The manager takeover date is **January 1, 2020**. The redesign decision uses information available through **December 31, 2019**, while the out-of-sample evaluation window runs from **2020-01-01 through 2025-12-31**.

Investment mandate. Build a long-horizon, growth-oriented retirement equity fund for an investor in their 30s. The objective is not to eliminate equity risk. The objective is to take growth risk intentionally, avoid unmanaged concentration, and improve risk-adjusted performance relative to both the inherited fund and SPY.

The final product is a dashboard-first financial analytics workflow backed by an Excel control workbook and a Python pipeline. The workbook stores assumptions, keep/drop/add decisions, candidate selections, constraints, and major outputs. Python performs the data processing, portfolio construction, backtesting, factor analysis, scenario analysis, and figure/table generation.

3 Data and Methodology

Daily adjusted price data are used to reconstruct the inherited fund, evaluate candidate stocks, and compare the revised portfolios. The S&P 500 Index is used as the benchmark and market proxy. The analysis compares four performance paths:

Table 1: Portfolio versions evaluated in the capstone.

Portfolio	**Description**
Legacy Fund	Original inherited fund continued after January 2020.
Revised Static Fund	Revised 10-stock portfolio, initialized in January 2020 and held without rebalancing.
Revised Active Fund	Same revised 10-stock portfolio, rebalanced monthly using the official active rule.
S&P 500 Index (SPY)	Benchmark for market comparison.

Performance is evaluated using terminal value, total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio, and maximum drawdown. Factor analysis is conducted using both full-sample CAPM and Fama–French 3-factor models, together with rolling 3-year factor exposures and stock-level FF3 contribution estimates for the final selected holdings. Scenario analysis includes historical realized stress tests (COVID crash day, COVID peak-to-trough, worst 21-day window, and worst 63-day window), deterministic portfolio shocks, factor stress scenarios, and a 1-year block-bootstrap terminal-return simulation.

4 Inherited Fund Diagnosis

The inherited fund performed well, but its risk profile changed materially over time. A portfolio that began as a 10-stock equal-weight fund became concentrated in a smaller group of winners.

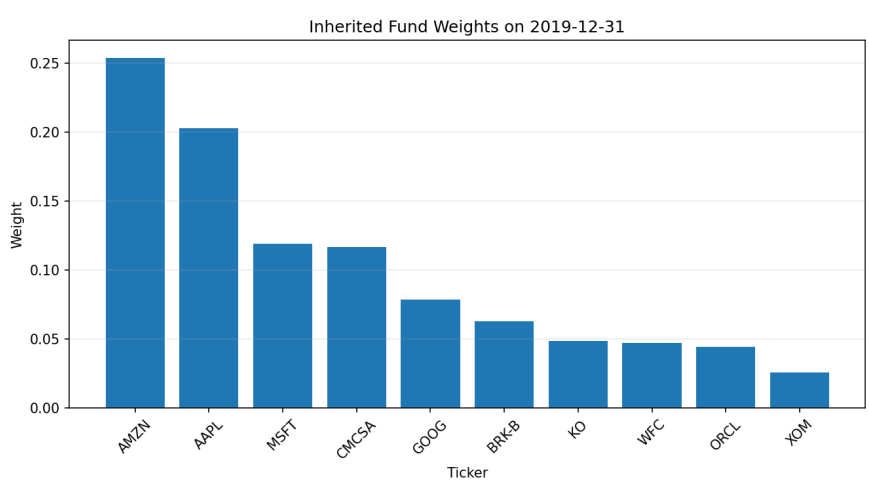

Figure 1: **Inherited fund weights as of the January 2020 decision date.** The original equal-weight structure has drifted into a winner-concentrated portfolio.

From Figure 1, by the decision date, AMZN and AAPL together accounted for a large share of the portfolio, and the top few holdings drove much of the fund's economic exposure. The key point is that the inherited fund was not broken. Its historical success came partly from allowing winners to run. However, for a new fund manager taking over in January 2020, the concentration creates a governance and risk-management problem. The fund needs a clearer mandate and a more deliberate implementation rule.

Table 2: Legacy holding decisions.

Legacy holding	**Decision**	**Rationale**
AAPL	Keep	Strong core compounder and important growth exposure.
AMZN	Keep	Major long-run winner with strong growth characteristics.
BRK-B	Keep	Provides diversification away from pure platform and technology exposure.
GOOG	Keep	High-quality platform business with long-run growth potential.
MSFT	Keep	Durable quality-growth holding with a strong role in the revised fund.
CMCSA	Drop	Less compelling strategic role in the revised growth mandate.
KO	Drop	Defensive, but less aligned with the more aggressive retirement-growth objective.
ORCL	Drop	Plausible technology holding, but lower conviction than the selected alternatives.
WFC	Drop	Weaker fit with the revised mandate.
XOM	Drop	Weaker fit with the revised growth-oriented equity mandate.

5 Candidate Search and Final Revised Portfolio

The candidate search focuses on high-quality growth and compounding businesses that could improve the inherited fund's long-run return potential while broadening its sources of return. The revised portfolio intentionally tilts more aggressively than a defensive retirement fund would, because the investor is assumed to be in their 30s with a long horizon.

Table 3: Candidate Search Decisions

Ticker	**Decision**	**Rationale**
LMT	Reject	Strong defense candidate and useful diversifier, but the final mandate shifted toward a more aggressive growth-oriented retirement portfolio, and other additions offered higher long-run upside.

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Table 3 continued from previous page

Ticker	Decision	Rationale
NOC	Reject	Credible aerospace/defense candidate, but less compelling than the selected growth additions for a long-horizon investor seeking stronger compounding potential.
GD	Reject	Reasonable defense/industrial option, but offered less attractive growth exposure than the final selected candidates.
JNJ	Reject	High-quality defensive healthcare name, but too conservative relative to the portfolio's final growth-oriented objective.
PG	Reject	Strong defensive consumer company, but weaker fit for an aggressive retirement-growth mandate and less attractive than the chosen additions.
UNH	Add	Adds healthcare growth exposure, improves sector diversification, and offers stronger upside potential than more defensive healthcare alternatives.
COST	Add	Adds a high-quality non-tech compounder with resilient demand, strong execution, and diversification away from inherited mega-cap technology exposure.
HON	Reject	Solid industrial-quality candidate, but less compelling than the selected growth-oriented additions in return potential and strategic role.
NVDA	Add	Adds semiconductor and accelerated-computing growth exposure with very strong long-run compounding potential consistent with the final aggressive-growth mandate.
AVGO	Add	Adds semiconductor and infrastructure-technology exposure with strong cash-generation and long-run compounding characteristics.
ADBE	Add	Adds a high-margin software compounder that broadens the portfolio's growth sleeve beyond the inherited platform names.
TMO	Reject	Attractive quality-growth healthcare candidate, but not selected because the final portfolio already achieved healthcare exposure through UNH and stronger pure growth exposure elsewhere.
V	Reject	Excellent business and strong long-run compounder, but excluded because the final portfolio prioritized semiconductors, software, healthcare, and consumer quality additions.

NVDA and AVGO add semiconductor and infrastructure-technology exposure. ADBE adds a high-margin software compounder outside the inherited mega-cap platform names. COST adds a high-quality non-tech consumer compounder, while UNH adds healthcare growth exposure and sector diversification. The revised fund begins as an equal-weight 10-stock portfolio.

To sum up, the revised portfolio keeps five inherited holdings: AAPL, AMZN, BRK-B, GOOG, and MSFT; drops five inherited holdings: CMCSA, KO, ORCL, WFC, and XOM; and adds five new holdings: NVDA, AVGO, ADBE, COST, and UNH.

6 Backtest Results

The out-of-sample comparison uses a common January 2020 starting value. The new summary table is designed for report writing: it avoids mixing raw growth-of-one series with manager-facing common-start values.

Table 4: Common-start out-of-sample backtest summary, 2020–2025.

Portfolio	Terminal	Ann. ret.	Ann. vol.	Sharpe	Sortino	Max DD
Legacy Fund	15.26	18.48%	24.34%	0.76	1.00	-31.38%
Revised Static Fund	36.21	36.90%	30.30%	1.22	1.68	-35.33%
Revised Active Fund	24.07	27.86%	25.28%	1.10	1.45	-31.25%
S&P 500 Index (SPY)	12.68	14.87%	20.75%	0.72	0.88	-33.72%

The revised portfolios substantially outperform both the Legacy Fund and the S&P 500 Index. The Revised Static Fund is the terminal-wealth winner. However, that result comes with a major implementation caveat: the static version allows winners to run without control, creating new concentration risk. The Revised Active Fund delivers lower terminal wealth than Static, but it still meaningfully outperforms Legacy and SPY while better controlling drift.

From Table 4, the Static Fund demonstrates the power of letting winners run. This can be attractive for a long-horizon investor, but it also recreates the same kind of concentration problem that motivated the redesign. From Figure 2, the Active Fund provides a more disciplined implementation by rebalancing monthly and keeping the portfolio closer to the intended 10-stock structure.

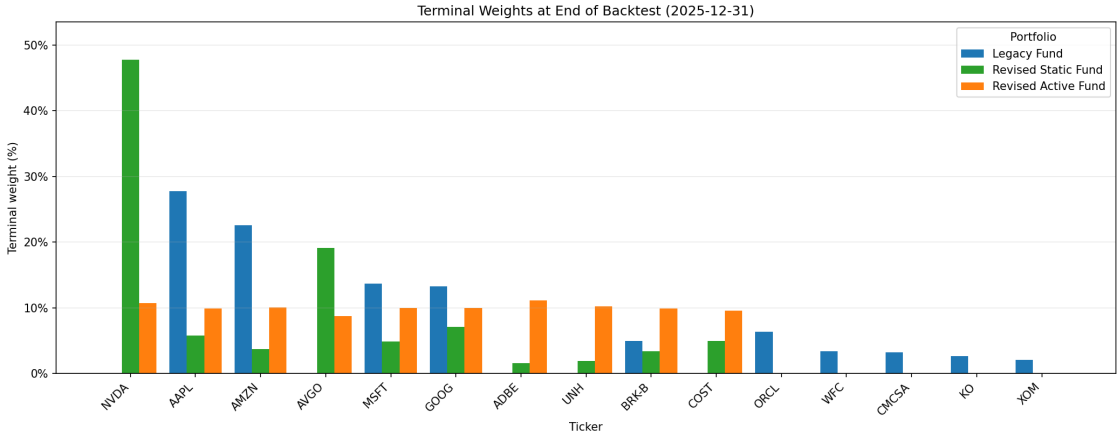

Figure 2: **Terminal portfolio weights on 2025-12-31.** The Legacy Fund remains concentrated in AAPL and AMZN, the Revised Static Fund becomes heavily concentrated in NVDA and AVGO, and the Revised Active Fund stays close to equal weight across the selected holdings.

The preferred recommendation is not simply the path with the highest historical terminal wealth. The preferred recommendation must also be implementable, transparent, and consistent with a retirement-oriented mandate. For that reason, the active implementation is favored despite the static portfolio's higher terminal value.

7 Active Rule Selection

The project tests three monthly active rebalancing rules. The **Equal Weight** is the simplest: at each monthly rebalance, it resets the selected 10-stock portfolio back to equal weights, so each holding returns to 10% of the portfolio. The **Balanced Growth** rule starts from those same static target weights but tilts them using inverse trailing volatility over the estimation window. In practice, lower-volatility stocks receive relatively larger weights, while higher-volatility stocks receive relatively smaller weights. The **Growth Tilt** rule also begins from the static target weights, but instead uses a Sharpe-style signal based on trailing average return divided by trailing volatility. Stocks with stronger positive risk-adjusted performance receive larger weights, while weaker names receive smaller weights. After the raw target weights are formed, they are normalized to sum to 100%, constrained by the minimum and maximum weight limits, and adjusted for the turnover cap before becoming effective on the next trading day after the month-end decision date.

Table 5: Active-rule comparison.

Active rule	Ann. ret.	Ann. vol.	Sharpe	Max DD	Avg. turnover
Equal Weight Baseline	27.84%	25.27%	1.10	-31.25%	2.50%
Balanced Growth	24.70%	23.54%	1.05	-28.66%	2.70%
Growth Tilt	28.03%	25.37%	1.10	-31.36%	2.32%

Although the **Growth Tilt** rule produces the highest annualized return, its advantage over the **Equal Weight** is very small. In the active-rule comparison, Growth Tilt earns an annualized return of about 28.03%, compared with 27.84% for Equal Weight, while both strategies deliver nearly identical Sharpe ratios of about 1.10. Growth Tilt also has slightly lower average turnover, but its maximum drawdown is marginally worse, and its added complexity does not materially improve the overall investment case. By contrast, the **Balanced Growth** rule reduces volatility and market beta, but it gives up too much return for the stated mandate of a long-horizon, growth-oriented retirement portfolio. Its annualized return falls to about 24.70%, with a Sharpe ratio of about 1.05, making it the most conservative of the three active rules.

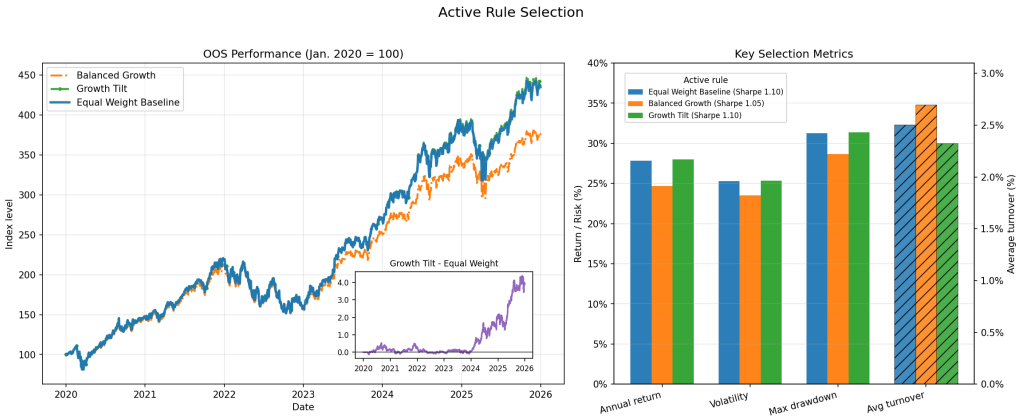

Figure 3: **Active rule comparison.** The left panel shows out-of-sample performance for the three active rules from January 2020 onward, and the inset highlights the small cumulative advantage of Growth Tilt over Equal Weight. The right panel compares annual return, volatility, max drawdown, and turnover.

For that reason, the central choice is really between Growth Tilt and Equal Weight. Since Equal Weight delivers nearly the same return and risk-adjusted performance while remaining much simpler, more transparent, and easier to explain, it is the better final implementation choice. In a portfolio-management setting, that simplicity matters: a rule that is easy to monitor and defend is often preferable to a more complicated rule that produces only a trivial improvement in backtest performance.

The official active rule is **monthly equal-weight rebalancing**. It captures nearly the same upside as Growth Tilt while being more transparent, easier to monitor, and less vulnerable to overfitting. This simplicity is valuable in a professional recommendation.

8 Factor Analysis: CAPM and Fama–French 3-Factor Model

To explain why the revised portfolios outperform the Legacy Fund, I estimate both a full-sample CAPM and a full-sample Fama–French 3-factor (FF3) model over the 2020–2025 out-of-sample period. I then examine rolling 3-year exposures to show how the portfolios' factor profiles evolve through time. Finally, I decompose the final selected 10 holdings using the FF3 model to identify which stocks contribute most to the portfolio's excess return.

8.1 Full-sample factor results

Table 6 shows that both revised portfolios have much stronger estimated alpha than the Legacy Fund. Under CAPM, the Legacy Fund has an annualized alpha of 3.5% and a beta of 1.03, while the Revised Active Fund improves to 10.3% alpha with a beta of 1.09. The Revised Static Fund has the strongest CAPM alpha at 16.8%, but also the highest beta at 1.21. The FF3 results tell a similar story. All three portfolios load positively on the market factor and negatively on both SMB and HML, indicating a large-cap growth tilt. This pattern is strongest for the Revised Static Fund, which has the largest market beta and the most negative HML loading.

Table 6: Full-sample CAPM and FF3 summary, 2020–2025

Portfolio	**CAPM α**	**CAPM β**	**FF3 α**	**MKT-RF**	**SMB**	**HML**	**R^2 (FF3)**
Legacy Fund	3.5%	1.03	2.9%	1.04	-0.25	-0.27	0.872
Revised Static Fund	16.8%	1.21	16.5%	1.21	-0.27	-0.50	0.830
Revised Active Fund	10.3%	1.09	9.8%	1.09	-0.24	-0.34	0.916

8.2 Rolling 3-year factor exposures

The rolling 3-year results reinforce the same point. Figure 4 shows that the Revised Static Fund becomes increasingly growth-oriented over time: its market beta trends upward, and its HML loading becomes much more negative by the end of the sample. The Revised Active Fund also remains growth-tilted, but its factor exposures are more stable and less extreme. The Legacy Fund stays comparatively moderate on all three dimensions. In other words, the static implementation achieves its superior terminal wealth partly by allowing a stronger large-growth factor profile to

build over time, while the active implementation captures much of the same benefit without letting exposures drift as far.

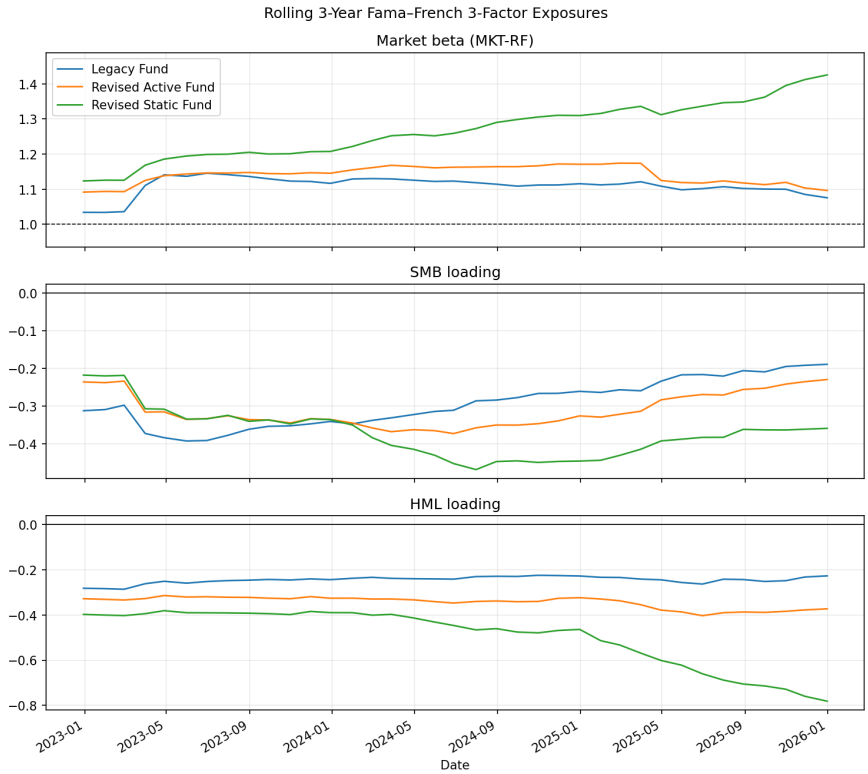

Figure 4: **Rolling 3-year Fama–French 3-factor exposures, updated monthly.** The Revised Static Fund becomes increasingly growth-tilted through time, while the Revised Active Fund remains growth-oriented but with more controlled factor drift.

8.3 Stock-level FF3 contribution

The stock-level FF3 decomposition in Table 7 and Figure 5 shows that NVDA and AVGO are the dominant contributors to the revised portfolio's excess return. NVDA has the highest annualized alpha contribution and the largest market-factor contribution, producing the strongest modeled excess return among the final 10 holdings. AVGO is the second-largest contributor. A second group of positive contributors includes GOOG, AAPL, MSFT, COST, and AMZN. By contrast, UNH and ADBE have much smaller modeled contributions, and both have negative alpha estimates even though they still contribute to portfolio diversification and long-run growth exposure.

Table 7: FF3 contribution of the final selected 10 holdings

Ticker	Alpha	Market	SMB	HML	Realized Excess Return
NVDA	46.2%	22.8%	0.4%	-0.6%	68.8%
AVGO	30.7%	18.6%	-0.1%	-0.2%	49.0%
GOOG	13.2%	14.6%	0.5%	-0.2%	28.0%

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Ticker	Alpha	Market	SMB	HML	Realized Excess Return
AAPL	8.2%	15.7%	0.7%	-0.2%	24.3%
MSFT	5.1%	15.1%	0.9%	-0.3%	20.8%
COST	9.9%	9.2%	0.7%	-0.2%	19.6%
AMZN	4.1%	14.4%	0.3%	-0.4%	18.4%
BRK-B	1.0%	10.7%	0.6%	0.3%	12.6%
UNH	-4.7%	10.4%	0.8%	0.1%	6.5%
ADBE	-10.5%	15.6%	0.4%	-0.4%	5.1%

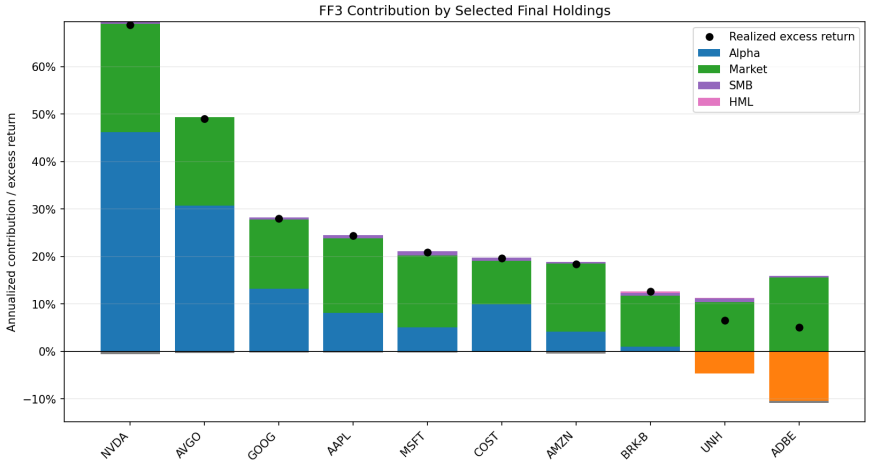

Figure 5: **FF3 contribution by the final selected holdings.** NVDA and AVGO are the dominant contributors to the revised portfolio's excess return, while the remaining holdings provide smaller but still meaningful modeled contributions.

Taken together, the factor results show that the revised portfolios outperform not only because of stock selection, but also because they shift the fund toward a stronger large-cap growth profile. The Revised Static Fund captures the greatest benefit from that shift, but it also develops the most extreme concentration and factor exposure. The Revised Active Fund preserves much of the alpha and growth tilt while keeping the implementation more disciplined. For that reason, the factor analysis supports the same conclusion as the backtest and terminal-weight results: the aggressive-growth revised portfolio is attractive, but the active implementation is the more balanced practical recommendation.

9 Scenario and Stress Testing

The revised portfolio is designed for a long-horizon, growth-oriented investor in their 30s, so the goal is not to eliminate equity risk. Instead, the goal is to understand where the revised portfolios remain vulnerable and whether active implementation improves robustness relative to the static version. To evaluate this, I examine four types of stress tests: historical realized stress events,

deterministic portfolio shocks, factor-based shocks, and a 1-year bootstrap distribution of terminal returns.

9.1 Historical realized stress tests

The historical stress tests show that the revised portfolios remain fully exposed to major market drawdowns, but they still compare favorably with the benchmark. On the COVID crash day (2020-03-16), the Legacy Fund loses about 9.9%, while the Revised Static and Revised Active Funds lose about 12.9% each, compared with 10.9% for SPY. The revised portfolios are therefore more sensitive than the Legacy Fund on that single day, which is consistent with their stronger growth orientation.

However, over the full COVID peak-to-trough window (2020-02-19 to 2020-03-23), both revised portfolios perform substantially better than SPY. The Legacy Fund falls about 25.0%, the Revised Static Fund about 25.5%, and the Revised Active Fund about 25.8%, while SPY falls about 33.4%. A similar pattern appears in the worst rolling 21-day window, where all three funds lose around 24–25%, compared with more than 32% for SPY. Over the worst 63-day window, the revised portfolios again hold up relatively well, with losses of about 12–13%, compared with 14.1% for the Legacy Fund and 23.8% for SPY.

These results indicate that the revised portfolios are not low-risk, but they are still more resilient than the benchmark over severe multi-week market stress periods.

9.2 Deterministic portfolio shocks

The deterministic shock tests show the clearest difference between the static and active implementations. Unsurprisingly, a parallel -10% or -20% shock produces the same impact for all portfolios. More interesting are the portfolio-specific shocks.

Under the **Growth / Duration Shock**, which applies a larger decline to growth-linked holdings, the Revised Static Fund loses about 18.8%, compared with 17.3% for the Legacy Fund and 16.4% for the Revised Active Fund. Under the **Semiconductor Shock**, where NVDA and AVGO fall 30% and other growth names fall 12%, the Revised Static Fund loses 23.3%, far worse than the Legacy Fund at 10.4% and the Revised Active Fund at 13.4%. The most informative test is the **Concentration Unwind**, in which the top two terminal-weight holdings in each portfolio fall 25% and all remaining holdings fall 5%. In this case, the Revised Static Fund loses 18.4%, the Legacy Fund loses 15.1%, and the Revised Active Fund loses only 9.4%.

These deterministic stress tests show that the static portfolio's superior wealth comes with substantial vulnerability to a reversal in its largest winning positions. The active version materially reduces that concentration risk.

9.3 Factor stress scenarios

The factor-based shocks reinforce the same conclusion. A **Market Selloff** of -10% produces losses of about 10.4% for the Legacy Fund, 12.1% for the Revised Static Fund, and 10.9% for the Revised Active Fund. Because the revised portfolios have stronger growth and market exposure, they are slightly more vulnerable than the Legacy Fund to a broad market decline.

A **Value Rotation** and **Small-Cap Rotation** also penalize the revised portfolios because they have more negative HML and SMB loadings than the Legacy Fund. These losses are relatively modest, but they confirm that the revised portfolios are more exposed to a reversal in the large-cap growth style that helped them outperform. The most severe factor scenario is the **Growth Squeeze**, which combines a market selloff with simultaneous value and small-cap outperformance. In this case, the Revised Static Fund loses about 18.4%, the Revised Active Fund about 15.9%, the Legacy Fund about 14.9%, and the benchmark about 12.0%.

This factor-stress analysis is especially important because it links the scenario section directly to the FF3 results: the revised portfolios outperform partly because they take on a stronger large-cap growth profile, and they are correspondingly more vulnerable if that factor environment reverses.

9.4 Bootstrap distribution of terminal returns

The 1-year block-bootstrap distribution compares the overall return profile of the static and active implementations. The Revised Static Fund has the higher mean and median terminal return, about 42.9% and 38.6%, respectively, compared with 31.0% and 29.2% for the Revised Active Fund. It also has a slightly lower probability of finishing with a loss, about 11.3% versus 12.9% for the Active Fund.

At the same time, the Active Fund has a somewhat better extreme downside profile. Its 1st percentile terminal return is about -24.3%, compared with -27.0% for the Static Fund, and its CVaR_{95} is about 19.9%, compared with 21.2% for the Static Fund. In other words, the static implementation offers more upside on average, but the active implementation produces somewhat less severe extreme-tail outcomes.

Taken together, the stress results support the same conclusion as the backtest and factor sections. The aggressive-growth revised portfolio is attractive because it improves terminal wealth and risk-adjusted performance relative to both the Legacy Fund and SPY, but it remains an equity portfolio with meaningful growth and concentration risk. The Revised Static Fund generates the highest wealth but is much more exposed to semiconductor and concentration-unwind shocks. The Revised Active Fund preserves much of the same upside while offering materially better protection against portfolio drift and concentrated winner reversals.

For that reason, the stress analysis supports the final recommendation to use the aggressive-growth revised portfolio, but to implement it with monthly equal-weight rebalancing rather than a static buy-and-hold structure.

10 Limitations

Several limitations should be acknowledged when interpreting the results of this project. First, the portfolio redesign is evaluated over a single realized out-of-sample period, 2020–2025, which includes unusually strong performance from large-cap growth, semiconductor, and software-related firms. As a result, the revised portfolio's outperformance may partly reflect a favorable realized market regime rather than a strategy that will necessarily dominate in future periods.

Second, the revised portfolio remains an aggressive equity portfolio with meaningful exposure to growth and technology-related factors. Although the active implementation reduces concentration

relative to the static version, the portfolio is still vulnerable to broad equity drawdowns, growth-stock derating, and semiconductor-specific stress. The scenario and stress tests show that the revised portfolios remain materially exposed to adverse factor reversals and concentration-driven shocks.

Third, the factor analysis is explanatory rather than predictive. The CAPM and Fama–French 3-factor regressions help explain the sources of realized performance, but they do not guarantee that the same alpha or factor relationships will persist going forward. In particular, the revised portfolios' stronger negative HML and SMB loadings indicate a large-cap growth tilt, which may underperform in different market environments.

Fourth, the backtests rely on simplified implementation assumptions. Transaction costs are modeled in a stylized way, and the analysis does not incorporate taxes, liquidity constraints, market impact, or retirement-specific cash-flow needs. These simplifications are reasonable for a course project, but they mean the reported results should be interpreted as strategy-level evidence rather than a fully realistic personal financial plan.

Finally, the static implementation illustrates an important practical limitation: a portfolio can generate very strong historical wealth while drifting into an undesirable concentration profile. This reinforces the importance of distinguishing between raw backtest success and a strategy that is actually suitable for implementation. For that reason, the final recommendation emphasizes not only return outcomes, but also concentration control, transparency, and repeatability.

11 Final Recommendation

The final recommendation is to replace the inherited fund with a revised 10-stock portfolio designed for a long-horizon investor in their 30s. The revised fund is intentionally growth-oriented rather than defensive. Its objective is to preserve the strongest inherited compounders, expand the portfolio's sources of return, and improve long-run risk-adjusted performance relative to both the Legacy Fund and the S&P 500 Index.

11.1 Theme of the revised fund

The revised portfolio is built around the following theme:

> *A growth-oriented retirement equity portfolio that preserves high-quality inherited winners while adding new long-run compounders in semiconductors, software, healthcare, and consumer quality.*

This theme reflects the view that a retirement investor in their 30s can tolerate meaningful equity risk, but that risk should be taken deliberately rather than through unmanaged concentration in a small number of inherited winners.

11.2 Keep, drop, and add decisions

The revised portfolio keeps five inherited holdings: **AAPL, AMZN, BRK-B, GOOG, and MSFT.**

These names remain because they are the strongest inherited compounders and continue to play central roles in the revised fund's long-run growth profile.

The revised portfolio drops five inherited holdings: **CMCSA, KO, ORCL, WFC, and XOM.**

These names are removed because they are less aligned with the final growth-oriented mandate or are less compelling than the selected replacement candidates.

The revised portfolio adds five new holdings: **NVDA, AVGO, ADBE, COST, and UNH.**

These additions strengthen the portfolio's growth and compounding profile while broadening its sources of return. **NVDA** and **AVGO** add semiconductor and infrastructure-technology exposure. **ADBE** adds high-margin software exposure. **COST** contributes a high-quality non-tech compounder, and **UNH** adds healthcare growth and sector diversification.

11.3 How the revised fund is implemented

The revised fund is initialized at equal target weights across the 10 selected holdings. The static version is useful as a reference because it shows the wealth potential of the chosen names, but it is not the preferred implementation. Over time, the static portfolio becomes highly concentrated in a small number of winners, especially NVDA and AVGO.

The preferred implementation is the **Revised Active Fund** using **monthly equal-weight rebalancing**. Under this rule, the portfolio is reset to equal weights at each monthly rebalance date, and the new weights take effect on the next trading day. This approach is selected because it captures much of the upside of the revised stock selection while preventing the fund from drifting into an extreme concentration profile.

11.4 Why this recommendation is preferred

The revised portfolio outperforms both the Legacy Fund and SPY over the 2020–2025 out-of-sample period. The static implementation produces the highest terminal wealth, but it does so by allowing extreme winner-driven concentration to form. The active implementation gives up some upside relative to the static version, but it still strongly outperforms the Legacy Fund and S&P 500 Index while keeping the portfolio much closer to its intended structure.

For that reason, the final recommendation is not simply to buy the revised 10 stocks and ignore them. The recommendation is to adopt the revised growth-oriented portfolio *and* implement it with monthly equal-weight rebalancing. This preserves the central strengths of the redesign — stronger long-run growth potential, improved risk-adjusted performance, and better diversification across business models — while reducing the concentration risk that would otherwise emerge over time.

Appendices

A Data and Sample Setup

This appendix documents the data sources, sample windows, benchmark definition, and portfolio construction timing used in the analysis.

A.1 Data sources

- Daily adjusted prices: Yahoo Finance / `yfinance`
- Benchmark: SPY
- Risk factors: Kenneth French daily Fama–French 3-factor data

A.2 Active implementation settings

- active rule: equal_weight
- estimation window: 36 months
- maximum weight: 25%
- minimum weight: 0%
- turnover cap: 20%
- transaction cost assumption: 10 bps

B Portfolio Construction Details

This appendix summarizes the final revised portfolio composition and the portfolio-weight outputs used in the backtests.

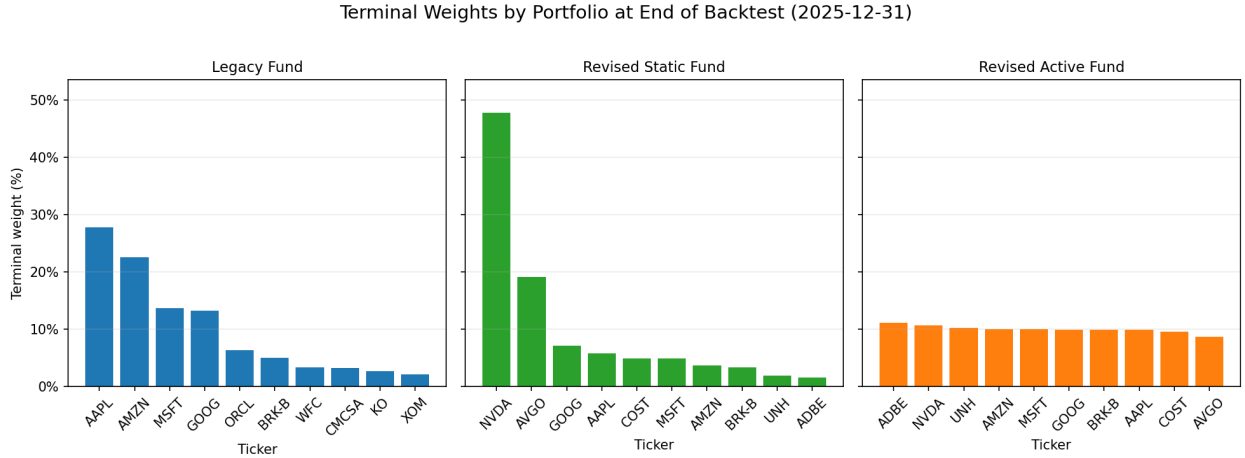

Figure B1: Terminal portfolio weights on 2025-12-31 by funds.

Table B1: Terminal weight comparison

Ticker	Legacy	Revised Static	Revised Active
AAPL	27.75%	5.73%	9.88%
ADBE	0.00%	1.60%	11.08%
AMZN	22.59%	3.72%	10.03%
AVGO	0.00%	19.07%	8.72%
BRK-B	4.97%	3.36%	9.92%
CMCSA	3.25%	0.00%	0.00%
COST	0.00%	4.95%	9.57%
GOOG	13.27%	7.07%	9.94%
KO	2.63%	0.00%	0.00%
MSFT	13.70%	4.85%	9.96%
NVDA	0.00%	47.74%	10.68%
ORCL	6.36%	0.00%	0.00%
UNH	0.00%	1.90%	10.21%
WFC	3.39%	0.00%	0.00%
XOM	2.08%	0.00%	0.00%

C Scenario and Stress Test Details

This appendix reports the full scenario and stress-test outputs supporting the risk discussion in the main text.

C.1 Historical realized stress tests

Table C1: Historical stress-test summary

Scenario	Legacy	Revised Static	Revised Active	SPY
COVID Crash Day	-9.86%	-12.89%	-12.91%	-10.94%
COVID Peak-to-Trough	-24.99%	-25.53%	-25.76%	-33.40%
Worst 21-Day Window	-23.57%	-24.38%	-24.64%	-32.75%
Worst 63-Day Window	-14.10%	-12.42%	-12.57%	-23.79%

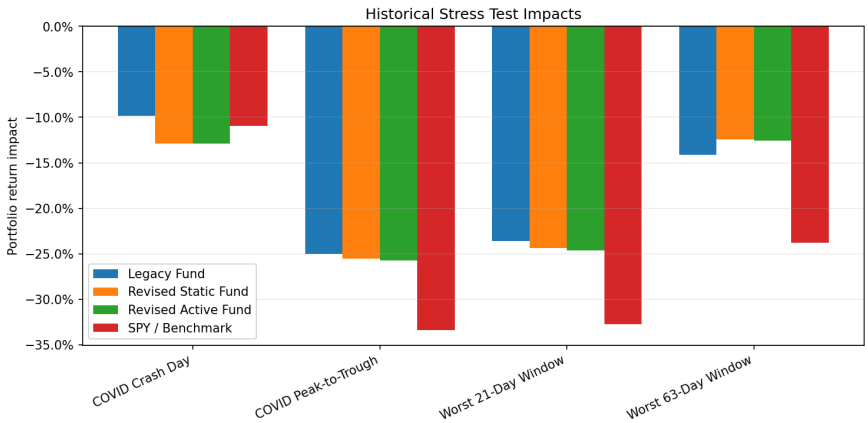

Figure C1: Historical realized stress-test impacts.

C.2 Deterministic portfolio shocks

Table C2: Deterministic portfolio shock summary

Scenario	Legacy	Revised Static	Revised Active	SPY
Parallel -10% Shock	-10.00%	-10.00%	-10.00%	-10.00%
Parallel -20% Shock	-20.00%	-20.00%	-20.00%	-20.00%
Growth / Duration Shock	-17.28%	-18.77%	-16.44%	—
Semiconductor Shock	-10.41%	-23.31%	-13.41%	—
Concentration Unwind	-15.07%	-18.36%	-9.35%	—

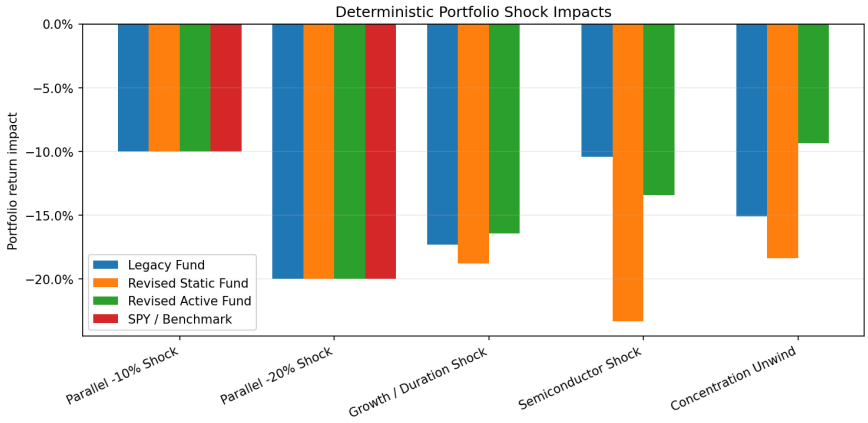

Figure C2: Deterministic portfolio shock impacts.

C.3 Factor stress scenarios

Table C3: Factor stress summary

Scenario	Legacy	Revised Static	Revised Active	SPY
Market Selloff	-10.43%	-12.12%	-10.94%	-10.00%
Value Rotation	-1.35%	-2.50%	-1.69%	0.00%
Small-Cap Rotation	-1.27%	-1.34%	-1.18%	0.00%
Growth Squeeze	-14.89%	-18.36%	-15.87%	-12.00%

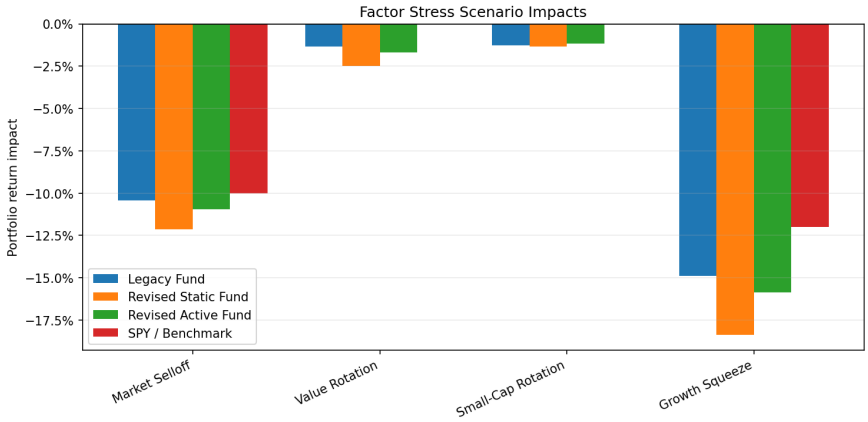

Figure C3: Factor stress-scenario impacts.

C.4 Bootstrap distribution

Table C4: Bootstrap terminal-return summary

Portfolio	Mean	Median	5th Pctl	$CVaR_{95}$	Prob. Loss
Revised Static Fund	42.92%	38.56%	-11.42%	21.19%	11.33%
Revised Active Fund	31.01%	29.19%	-11.16%	19.87%	12.87%

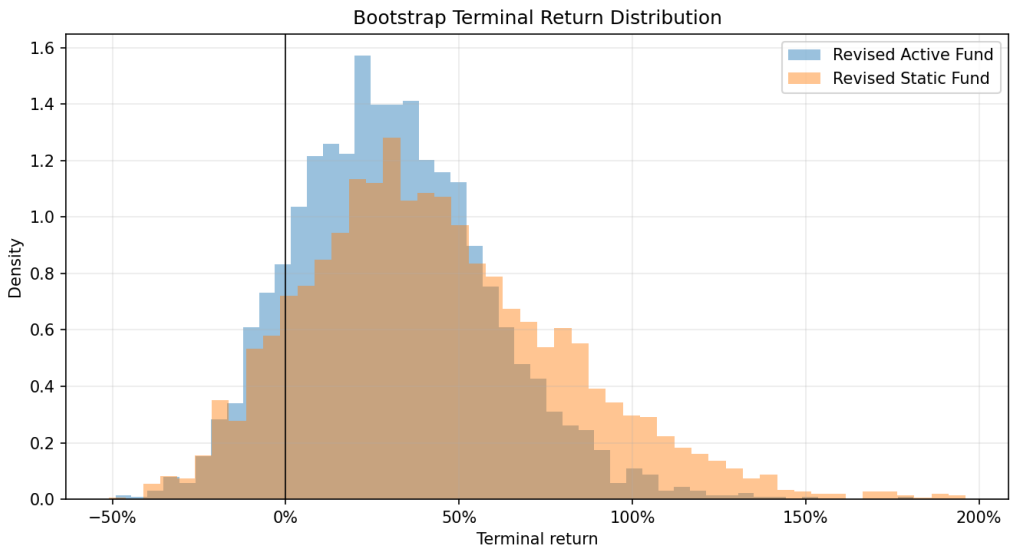

Figure C4: Bootstrap distribution of 1-year terminal returns.

D Reproducibility Note

The project is implemented through a reproducible workflow that combines an Excel control workbook, a Python analytics pipeline, and a local dashboard. The workbook stores assumptions, final portfolio decisions, and active-rule settings. The Python pipeline reads these inputs, generates the cleaned data, backtest outputs, factor-model results, and scenario/stress results, and saves the final tables and figures used in both the dashboard and the written report. This structure ensures that the results presented in the report and presentation are generated from the same underlying workflow rather than being manually re-created.